

TRILOKESH RANJAN SARKAR

AI/ML Engineer | GenAI & LLM Automation | Computer Vision | NLP | 2 Yrs Experience

8910384107 | trilokesh086@gmail.com | LinkedIn | GitHub | India | arXiv

SUMMARY

AI/ML Engineer transitioning from a **Data Analyst** role, with 2 years building **LLM-powered automation systems, OCR/NLP pipelines, and computer vision solutions** across Retail, Healthcare, and BFSI. Experienced designing production workflows that combine **LLM reasoning, tool-calling** to external data sources, **human-in-the-loop validation**, and **automated error-correction** — reducing manual effort by up to 80% and improving operational scalability. Strong foundation in **adversarial machine learning** and **model robustness**, evidenced by a peer-reviewed **arXiv publication**. Comfortable across **Python, SQL**, computer vision (**YOLOv8**), Generative AI (**GPT-4o**), **NLP**, and cloud data infrastructure (**AWS, Redshift**). Seeking **AI/ML Engineer or Data Scientist** roles building production LLM and automation systems.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Regex

- **AI/ML:** Machine Learning, Artificial Intelligence, Deep Learning, Computer Vision, Image Classification, Transfer Learning, Model Evaluation & Benchmarking, API Integration, Workflow Automation
- **Generative AI & LLMs:** GPT-4o, GPT-4o API, Prompt Engineering, LLM-driven Workflow Design, Tool Calling, Human-in-the-Loop (HITL) Validation, LLM Output Evaluation, Resource-Constrained Fine-Tuning
- **NLP & Document AI:** Named Entity Recognition (NER), Intelligent Text Parsing, OCR/NLP Automation, Py-Tesseract OCR, Regex-based Extraction, Document Digitization, Data Annotation
- **Adversarial ML & Model Security:** Adversarial Robustness Testing, FGSM, Carlini-Wagner (CW) Attacks, Defensive Distillation, Teacher-Student Model Distillation, CNN Architectures (ResNeXt-50, DenseNet-201, VGG-19, ResNet-101)
- **Data Engineering & Cloud:** AWS, Amazon Redshift, PostgreSQL, Apache Airflow, ETL Pipelines, Label Studio
- **BI & Visualization:** Power BI, Looker Studio, Interactive Dashboarding, KPI Reporting
- **Datasets & Benchmarks:** Tiny-ImageNet, CIFAR-10

EXPERIENCE

Data Analyst

10/2024 – Present

Web Spiders | Kolkata, West Bengal

- **LLM-powered automation pipeline:** Designed and orchestrated an LLM-driven validation system with **GPT-4o API integration, tool-calling** to external data sources, automated web research, and **human-in-the-loop review checkpoints** — built as a **large-scale workflow automation** pipeline on **Apache Airflow, AWS, Redshift, PostgreSQL**, and **Label Studio** — to automate **retail taxonomy validation** and **ontology auditing**, improving **data quality** and governance at scale.
- **Structured extraction with validation routing:** Built **OCR/NLP-based extraction pipelines** with **confidence-based routing** to human review, using **Named Entity Recognition (NER)** and **intelligent text parsing** for **document digitization** of handwritten and scanned documents — improving data accuracy and reducing manual effort by 80%.
- **Handwriting OCR pipeline:** Built a custom **Py-Tesseract**-based handwriting OCR pipeline for **document digitization** of handwritten medical documents, layering **NLP-based NER** and **regex matching** to cut human intervention in data extraction by 80%.
- **Computer vision — ANPR system:** Developed an **Automatic Number Plate Recognition (ANPR)** system using **YOLOv8** and **deep learning-based computer vision** for **real-time**, high-accuracy license plate detection and recognition.
- **Computer vision — in-store analytics:** Deployed **pre-trained machine learning models** for **real-time sentiment analysis** of in-store customer facial expressions and built **heat maps** from **video analytics** to identify high-traffic areas and peak hours, informing store layout and staffing decisions.
- **Analytics & BI:** Used **SQL, Python**, and **Power BI** for **business intelligence**, analysing trends, building **interactive dashboards**, and generating actionable insights for **data-driven decision-making**; built marketing analytics dashboards tracking email campaign **KPIs** that improved engagement 15% and conversions 12%.

Tech Stack: Python, SQL, GPT-4o API, Power BI, AWS, Apache Airflow, YOLOv8, Machine Learning, Regex, NLP

- **Adversarial robustness research:** Evaluated the security of a **Generative AI** model against **adversarial threats** by comparing **FGSM** and **Carlini-Wagner attacks** and assessed **distillation** as a defence mechanism.
- **Model fine-tuning under constraints:** Improved **model robustness** and efficiency through optimized **LLM fine-tuning** under limited compute resources.

PUBLICATION

Comparing FGSM, Carlini-Wagner Attacks, and the Function of Distillation as a Defence Mechanism to Assess Adversarial Robustness

arXiv.org — 2023–2024 | <https://arxiv.org/abs/2404.04245>

- **Scope:** Technical study of **adversarial attacks** on **deep neural networks** used for **image classification**, evaluating **FGSM** and **Carlini-Wagner (CW) attacks** against three pre-trained classifiers — **ResNeXt-50**, **DenseNet-201**, and **VGG-19** — on the **Tiny-ImageNet** dataset.
- **Defence mechanism:** Proposed and evaluated **defensive distillation** as a countermeasure on **CIFAR-10**, using **ResNet-101** as the teacher model and **ResNeXt-50** as the student model.
- **Findings:** Showed the **distillation-based defence** was effective against **FGSM** but remained partially vulnerable to the more sophisticated **CW attack**, with results validated through rigorous experimentation.
- **Recognition:** **Peer-reviewed** after review by **GenAI researchers**; contributes an **evaluation framework** comparing model vulnerabilities and defence effectiveness, offering actionable insights for **secure model deployment**.

EDUCATION

Post-Graduation Program in Data Science — Praxis Business School

2023 – 2024 | Kolkata, West Bengal | CGPI: 6.2 / 8

Bachelor of Computer Application — B. P. Poddar Institute of Management and Technology

2020 – 2023 | Kolkata, West Bengal | CGPA: 8.8 / 10

CERTIFICATIONS

- Databricks Fundamentals Accreditation — Databricks
- Generative AI Fundamentals Accreditation — Databricks
- SQL Certificate — Hacker Rank